MUTUAL FUND ANALYTICS PLATFORM

Final Project Report

Submitted By:

K S KEERTHANA

Organization:

Bluestock Fintech

Internship Duration:

7 Days

ACKNOWLEDGEMENT

I sincerely express my gratitude to Bluestock Fintech for providing me the opportunity to work on the Mutual Fund Analytics Platform. I am thankful to the mentors and team members whose guidance and support helped me successfully complete this internship project.

I also thank the mentors and team members who guided me throughout the internship. This project helped me understand data analysis, financial analytics, and dashboard development using modern tools.

ABSTRACT

The Mutual Fund Analytics Platform is an end-to-end financial analytics solution developed to evaluate mutual fund performance, investor behavior, portfolio risk, and market trends. The project integrates data preprocessing, exploratory data analysis, performance evaluation, risk analytics, and interactive dashboard development using Python, SQLite, and Power BI.

The project combines data cleaning, exploratory analysis, performance measurement, risk analytics, and dashboard visualization.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used for analysis, while Power BI was used for dashboard development.

CHAPTER 1: INTRODUCTION

Mutual funds have become one of the most popular investment options in India. Investors require proper analysis tools to evaluate risk, return, and market behavior.

The purpose of this project is to build a complete analytics platform that helps users understand mutual fund performance and investor trends.

CHAPTER 2: PROBLEM STATEMENT

Investors often face difficulties in:

- Comparing multiple funds.
- Understanding risk levels.
- Tracking SIP growth.
- Analyzing market performance.
- Making investment decisions.

This project attempts to solve these problems using data analytics techniques.

CHAPTER 3: OBJECTIVES

The main objectives are:

1. Data cleaning and preprocessing.
2. Database creation.
3. Exploratory data analysis.
4. Performance analytics.
5. Risk analysis.
6. Dashboard development.
7. Final reporting.

CHAPTER 4: TOOLS AND TECHNOLOGIES

Tool	Purpose
Python	Data analysis
Pandas	Data manipulation
NumPy	Numerical calculations
Matplotlib	Charts
Seaborn	Statistical visualization
SQLite	Database
Jupyter Notebook	Analysis
Power BI	Dashboard
GitHub	Version control

CHAPTER 5: DATASET DESCRIPTION

The project uses:

- Fund master data
- NAV history
- AUM information
- SIP inflows
- Investor transactions
- Portfolio holdings
- Benchmark indices

The datasets contain information about fund performance, investors, and market behavior.

CHAPTER 6: DATA CLEANING

Several preprocessing steps were performed:

- Missing value handling
- Duplicate removal
- Date conversion
- Data validation
- Standardization of transaction types
- Verification of numerical values

The cleaned data was stored in processed CSV files.

CHAPTER 7: DATABASE DESIGN

SQLite database was created.

Tables:

- dim_fund
- dim_date
- fact_nav
- fact_transactions
- fact_aum
- fact_performance

Relationships were established using primary and foreign keys.

CHAPTER 8: EXPLORATORY DATA ANALYSIS

The following analyses were performed:

- NAV trend analysis
- SIP growth analysis
- Category inflow analysis
- Investor demographics
- Geographic distribution
- Sector allocation
- Correlation analysis

Multiple charts and visualizations were generated.

CHAPTER 9: PERFORMANCE ANALYTICS

The following metrics were calculated:

- Annual Return
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

These metrics help evaluate fund performance.

CHAPTER 10: ADVANCED ANALYTICS

Advanced analytics includes:

- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Rolling Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Sector Concentration

These methods provide deeper insights into risk and investment behavior.

CHAPTER 11: DASHBOARD DEVELOPMENT

Four Power BI pages were developed:

Industry Overview

AUM, SIP inflows, fund houses.

Fund Performance

Returns, risk, scorecards.

Investor Analytics

Investor demographics and transactions.

SIP Trends

Market trends and inflows.

CHAPTER 12: KEY FINDINGS

1. SIP inflows increased continuously.
2. Equity funds generated higher returns.
3. Younger investors preferred SIP investments.
4. Large fund houses dominated AUM.
5. Risk-adjusted performance varied among funds.
6. Portfolio diversification reduced risk.
7. Some funds experienced larger drawdowns.
8. Market corrections impacted performance.

CHAPTER 13: CHALLENGES FACED

During the project, several challenges were encountered:

- File path errors.
- Missing data handling.
- Database design issues.
- Power BI learning difficulties.
- Git and GitHub problems.
- Understanding financial formulas.

These challenges were solved through practice and experimentation.

CHAPTER 14: LEARNINGS FROM THE PROJECT

This project helped me learn:

- Python programming.
- Pandas and NumPy.

- Data cleaning techniques.
- SQL and SQLite.
- Power BI dashboard development.
- Git and GitHub.
- Financial analytics concepts.
- Risk measurement techniques.

CHAPTER 15: LIMITATIONS

- Real-time data was not available.
- Some investor data was simulated.
- Limited historical period.
- Dashboard was developed locally.

CHAPTER 16: CONCLUSION

The Mutual Fund Analytics Platform demonstrates the practical application of data analytics in the financial domain by integrating ETL processes, database management, exploratory analysis, risk measurement, and interactive business intelligence dashboards. This project strengthened my technical, analytical, and problem-solving skills while providing valuable hands-on industry experience.

The project integrates ETL, analytics, risk measurement, dashboard development, and reporting into a single platform.

As a beginner in data analytics, this project provided practical experience in real-world tools and improved technical and analytical skills.

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4. NumPy Documentation
5. Power BI Documentation
6. SQLite Documentation
7. Financial market reports

DECLARATION

I hereby declare that the work presented in this report is my original work completed during the Bluestock Fintech internship.

Signature:

K S KEERTHANA